

A Two-Stage Density Collapse Quantum Testing Algorithm for Sophie Germain Primes

M.Chowdhury

Written 18th June 2025* Revised 29th September 2026*

Draft version (Dates of server time stamps)*

Abstract

Sophie Germain primes (SGPs) are conjectured to be infinite in number. We study our new bounded-gap conjecture (BGC) stating that every integer x has an SGP within forward distance $\lceil 8 \ln x \rceil^2$, together with a window-level relaxation (BGC-Macroscopic) that tolerates polynomially many uncovered integers in a window of 2^α integers. Our new conjectures imply the well known conjecture that there are infinite SGPs. We give a *fixed-interval tester* for a single window, with two stages. Stage 1 is a screen: a Deutsch–Jozsa-style circuit on a boosted SGP indicator, repeated $\text{poly}(\alpha)$ times. We prove that a single shot returns the all-zero string with probability exactly h^2 , where h is the fraction of uncovered odd offsets in the window. Hence the screen always flags a total SGP desert, almost never flags a window satisfying BGC-Macroscopic, and is blind to sparse violations. Stage 2 refines every flagged window with quantum counting, using $O(2^{\alpha/2})$ oracle calls to decide whether the window is empty. Regarding each quantum component of our new algorithm: Stage 1 has no advantage over classical random sampling (which is in fact more sensitive to sparse holes), so the quantum advantage of the tester is the quadratic speedup of Stage 2 in certifying that a flagged window is truly empty. The outputs are empirical evidence about the conjecture at the tested depth; they are not proofs of infinitude or of finiteness. We introduce a new specific way to boost probability of a solution

being read in Grover’s algorithm. Lemma 1 and 2, which is required to make our new algorithm work.

1 Introduction

A Sophie Germain prime (SGP) is a prime p such that $2p + 1$ is also prime. The SGP conjecture states that there are infinitely many. Cramér’s heuristic model treats the primality of n as an independent event of probability $1/\ln(n)$, so the probability that both n and $2n + 1$ are prime is approximately

$$\frac{1}{\ln(n)} \cdot \frac{1}{\ln(2n + 1)} \approx \frac{1}{\ln^2(n)}, \quad (3)$$

and the expected number of SGPs up to x is modeled by

$$\sum_{n=2}^x \frac{1}{\ln^2(n)} \sim \text{li}_2(x) \rightarrow \infty. \quad (2)$$

This is the standard heuristic reason to expect infinitely many SGPs; it is not a proof.

The Bounded Gap Conjecture (BGC)

We put forward the following conjecture, which is a new original empirical conjecture .

Conjecture 1 (BGC). For every integer $x \geq 2$ there is a Sophie Germain prime p with

$$x \leq p \leq x + \lceil 8 \ln x \rceil^2. \quad (1)$$

This covering form is equivalent, up to negligible rounding, to the gap form $SGP_{n+1} - SGP_n \leq \lceil 8 \ln(SGP_n) \rceil^2$ for all n . We use the covering form for two reasons. First, no SGP index appears: the bound is evaluated at the integer x being examined, which is exactly what an oracle can compute. Second, it is non-vacuous. A gap form quantified over “all n ” is satisfied vacuously by a finite sequence, whereas the covering form asserts the existence of an SGP ahead of *every* x .

Observation 1. If BGC holds then there are infinitely many SGPs, since every x has an SGP at or above it.

Consequently the tester’s outputs are not independent evidence for infinitude: any evidence for BGC is evidence for a statement that already implies infinitude.

Numerical support and a heuristic caveat. A direct sieve computation confirms Conjecture 1 for every x up to about $2 \cdot 10^7$. The largest forward distance to the next SGP observed in that range is about $8.1 \ln^2 x$, roughly an eighth of the allowed $64 \ln^2 x$. The same Cramér-type model that motivates the conjecture also suggests that its constants cannot survive to arbitrary scale. With the Hardy–Littlewood density $2C_2/\ln^2 x$ for SGPs ($C_2 \approx 0.66$), the average gap is about $0.76 \ln^2 x$, and the probability that a given x has no SGP within $64 \ln^2 x$ is about $e^{-84} \approx 2 \cdot 10^{-37}$, independently of the scale. Maximal gaps up to X then grow like $0.76 \ln^2 X \cdot \ln(X/\ln^2 X)$, which exceeds $64 \ln^2 X$ once $\ln X \gtrsim 84$, that is, for X of order 10^{36} and beyond. So the universal statement is heuristically expected to fail somewhere at astronomically large scales, which may be expressed with empirical evidence as true or false with out new quantum algorithm, although it holds with a wide margin in the checked range. These are heuristic estimates, not theorems. They motivate the window-level relaxation below (BGC-Macroscopic), which tolerates a small number of uncovered integers.

What this paper does

BGC is a statement about all integers, and no finite computation can establish it. What a finite computation can do is examine a window of $N = 2^\alpha$ consecutive integers deep in the number line and look for *density collapse*: a window in which the covering behaviour of BGC fails macroscopically, in the extreme case a window containing no SGP at all (an SGP desert). We give a modular *fixed-interval tester* that does this for one window, and can be run over successive windows to sweep a long range. The tester is a filter-and-refine pipeline: a cheap probabilistic screen (Stage 1), followed by a quantum-counting refinement (Stage 2) applied only to windows the screen flags.

Relation to prior work

The Deutsch–Jozsa literature applies the algorithm almost exclusively to Boolean oracles constructed to be constant or balanced. Our Stage 1 operates outside the exact promise regime, but on an oracle built from a number-theoretic object. In other words this paper uses the DJ oracle in a modified form and in a new way applied in relation to the SGP conjecture in number theory (which is not a toy problem). We are not aware of any prior work where the DJ oracle is used for a non-toy problem. We make no claim of quantum advantage for Stage 1 (Section 4). on the other hand the two quantum subalgorithm, used in both stages, components in our algorithm, together, give a conceptually new quantum only way to test the well known SGP conjecture giving a quantum advantage.

2 The Boosted Oracle

Lemma 1 (Boosted Oracle Window). Let $Q'(x)$ be an oracle indicating an isolated solution on $[0, 2^W - 1]$. Define the boosted oracle via a chain of logical ORs,

$$Q(x) = U\left(Q'(x) \vee Q'((x+1) \bmod 2^W) \vee \dots \vee Q'((x+\text{boost}(x)) \bmod 2^W)\right), \quad (4)$$

where $U(X) = 1$ if $X > 0$ and $U(X) = 0$ otherwise. Equivalently

$$Q(x) = U\left(\sum_{i=0}^{\text{boost}(x)} Q'((x+i) \bmod 2^W)\right) \quad (24)$$

. If a single real solution lies within $\text{boost}(x)$ steps ahead of x , then $Q(x) = 1$: one real solution makes every x in a run of length boost evaluate to 1 (spurious solutions).

For the SGP application we take $\text{boost}(y) = \lceil 8 \ln y \rceil^2$, the BGC bound evaluated at the integer y being examined.

Lemma 2 (Generalized Scattered-Coverage Oracle)

Nothing in the amplification argument requires the offsets to be consecutive, or their number to be logarithmic. Let $s : \mathbb{N} \rightarrow \mathbb{N}$ be any *spurious-solution*

generator producing offsets within an interval of width W , and let $m(x)$ be any bound on the number of offsets used. Define

$$Q(x) = U\left(Q'((x + s(0)) \bmod 2^W) \vee Q'((x + s(1)) \bmod 2^W) \vee \dots \vee Q'((x + s(m(x))) \bmod 2^W)\right), \quad (5)$$

with U as in Lemma 1. If a real solution lies at one of the scattered positions $x + s(i)$, then $Q(x) = 1$. Lemma 1 is the special case $s(i) = i$, $m(x) = \text{boost}(x)$. Lemma 2 is not used by the tester below; it is recorded as a general-purpose primitive for other constructions that need artificial local coverage around a real solution without contiguity or logarithmic size.

A concrete scattering rule. Fix an irrational γ , for instance $\gamma = \sqrt{p}$ for a chosen prime p , and set

$$s(i) = \left\lfloor ((i \cdot \gamma) \bmod 1) \cdot W \right\rfloor. \quad (6)$$

By the Weyl equidistribution theorem the sequence $\{i\gamma \bmod 1\}$ is equidistributed in $[0, 1)$, so the offsets $s(0), s(1), \dots$ spread quasi-uniformly over $[0, W)$ instead of forming a contiguous block; $m(x)$ is a free parameter.

The SGP indicators

Let $P(y)$ indicate primality, computed by a reversible primality test (Miller–Rabin or AKS). Define

$$S'(y) = P(y) \cdot P(2y+1), \quad S(y) = u\left(S'(y) \vee S'(y+1) \vee \dots \vee S'(y + \lceil 8 \ln y \rceil^2)\right), \quad (7)$$

with $u(z) = 1$ if $z > 0$ and $u(0) = 0$. Thus $S(y) = 1$ exactly when an SGP lies in $[y, y + \lceil 8 \ln y \rceil^2]$. Note that S is defined on all integers $y \geq 2$.

Why the logarithmic form of the bound matters

Because $\text{boost}(y) = \lceil 8 \ln y \rceil^2 = O(\log^2 y)$, the oracle for S is a chain of only $O(\log^2 y)$ OR'd evaluations of the primality test. This is a polynomial-size, efficiently constructible reversible circuit; a gap bound growing polynomially in y would make the oracle infeasible at the scale of an exponential window.

One evaluation of S costs $O(\log^2 y)$ primality tests, which we denote T_S (a polynomial in $\log y$). This factor multiplies every query count below and does not affect the comparisons.

3 Windows, Coverage, and BGC-Macroscopic

Definition (Window). For an integer start point $c \geq 2$ and a size parameter $\alpha \geq 1$, the *window* $W(c, \alpha)$ is the set of $N = 2^\alpha$ consecutive integers

$$W(c, \alpha) = \{c, c + 1, \dots, c + N - 1\}. \quad (10)$$

Its points are indexed by the *offset* $x \in \{0, \dots, N - 1\}$ via the integer $c + x$; the offset register is the α input qubits of the circuit in Section 4. The indicator S is not clipped to the window: evaluating $S(c + x)$ near the right end may inspect up to $\lceil 8 \ln(c + x) \rceil^2$ integers beyond $c + N - 1$.

Definition (Covered and uncovered integers). An integer $y \geq 2$ is *covered* if $S(y) = 1$, that is, if an SGP lies within forward distance $\lceil 8 \ln y \rceil^2$ of y , and *uncovered* otherwise. For a window let

$$U(c, \alpha) = \#\{x : S(c + x) = 0\} \quad (11)$$

be the number of uncovered integers in it. A window is a *desert* if $U(c, \alpha) = N$, i.e. every integer in it is uncovered. Since the intervals $[y, y + \lceil 8 \ln y \rceil^2]$ for consecutive y overlap, this holds exactly when no SGP lies in $[c, c + N - 1 + \lceil 8 \ln(c + N - 1) \rceil^2]$; in particular a desert contains, and is followed by, an SGP-free run of more than N integers.

BGC (Conjecture 1) says $U(c, \alpha) = 0$ for every window. We introduce a relaxation matched to what a probabilistic screen can resolve.

Conjecture 2 (BGC-Macroscopic). There is a polynomial q such that for all $\alpha \geq 1$ and all $c \geq 2$,

$$U(c, \alpha) \leq q(\alpha). \quad (12)$$

That is, every window of 2^α integers is covered by boosted SGPs except for at most polynomially many (in α , i.e. polylogarithmically many in N) uncovered integers.

BGC implies BGC-Macroscopic (take $q = 0$). The converse fails, and BGC-Macroscopic is the weaker statement that the heuristic caveat of Section 1 leaves plausible for window sizes up to roughly $2^{120} - 2^{140}$: under the

model there, the expected number of uncovered integers in a window of size N is about $2 \cdot 10^{-37}N$.

4 Stage 1: The Deutsch–Jozsa Screen/Filter

The circuit

Fix a window $W(c, \alpha)$ and set $n = \alpha$. Define

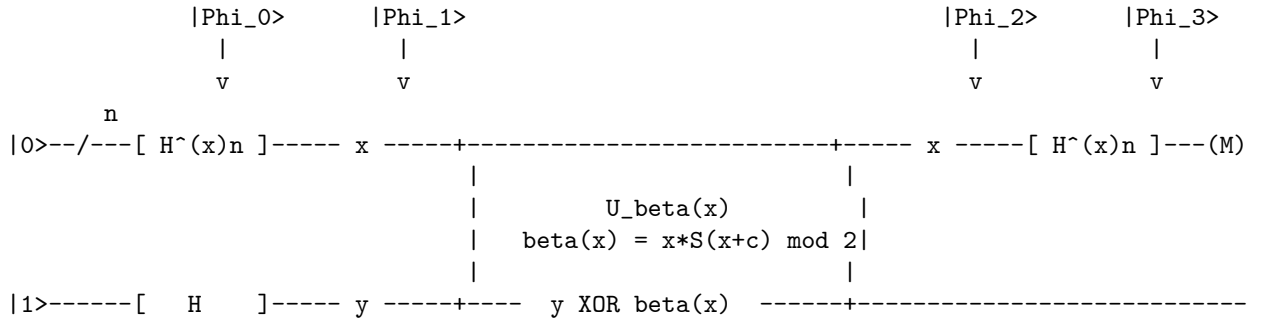
$$\beta(x) = x \cdot S(c + x) \pmod{2}, \quad (13)$$

so $\beta(x) = 0$ for even offsets x and $\beta(x) = S(c + x)$ for odd x . One shot of the screen is:

1. Initialize the n -qubit register to $|0\rangle^{\otimes n}$ and the ancilla to $|1\rangle$. Apply $H^{\otimes n} \otimes H$.
2. Apply the oracle $U_\beta : |x\rangle|y\rangle \mapsto |x\rangle|y \oplus \beta(x)\rangle$.
3. Apply $H^{\otimes n}$ to the input register and measure in the computational basis.

A shot is called *zero* if the outcome is $|0\rangle^{\otimes n}$ and *nonzero* otherwise. The circuit is drawn on the next page.

Circuit Diagram



(Figure 1: Stage-1 screen circuit for one window.)

Single-shot statistics

Proposition 1. Let u be the number of odd offsets $x \in \{0, \dots, N-1\}$ with $S(c+x) = 0$ (uncovered odd offsets), and let

$$h = \frac{u}{N/2} \in [0, 1]. \quad (14)$$

A single shot of the circuit returns $|0\rangle^{\otimes n}$ with probability exactly h^2 , and returns some other string with probability $1 - h^2$.

Proof. After the initial Hadamards the state is

$$|\Phi_1\rangle = \frac{1}{\sqrt{2^{n+1}}} \sum_{x=0}^{2^n-1} |x\rangle (|0\rangle - |1\rangle). \quad (15)$$

The oracle gives

$$|\Phi_2\rangle = \frac{1}{\sqrt{2^{n+1}}} \sum_{x=0}^{2^n-1} (-1)^{\beta(x)} |x\rangle (|0\rangle - |1\rangle). \quad (16)$$

The ancilla is unentangled and is dropped. The final $H^{\otimes n}$ maps $|x\rangle \mapsto 2^{-n/2} \sum_y (-1)^{x \cdot y} |y\rangle$, so

$$|\Phi_3\rangle = \sum_y \left[\frac{1}{2^n} \sum_x (-1)^{\beta(x) + x \cdot y} \right] |y\rangle, \quad (17)$$

and the amplitude on $y = 0$ is $a = \frac{1}{N} \sum_x (-1)^{\beta(x)}$. Even offsets contribute +1 each, a total of $N/2$. An odd offset contributes +1 if uncovered and -1 if covered, so the odd offsets contribute $u - (N/2 - u) = 2u - N/2$. Hence $\sum_x (-1)^{\beta(x)} = N/2 + 2u - N/2 = 2u$, and $a = 2u/N = h$. By the Born rule the probability of the all-zero outcome is $|a|^2 = h^2$. ■

(We verified the formula by direct simulation: for $\alpha = 10$, a single uncovered odd offset gives $a = 2/N$; a run of 40 uncovered consecutive offsets gives $h = 20/512$ and probability $h^2 \approx 1.5 \times 10^{-3}$, exactly as predicted.)

Consequences

1 and 3 below are "inherited pure cases" from the original D-J algorithm the other consequences 2,4,5 are the new outputs considered from the D-J algorithm once the original D-J promise(given in Deutsch-Josza 1992 paper) is dropped.

1. **A desert is always flagged.** If every integer of the window is uncovered, then $h = 1$ and every shot is zero, with certainty.
2. **A nonzero shot certifies non-emptiness.** A nonzero outcome occurs with positive probability only if $h < 1$, i.e. only if some odd offset is covered. So in the noiseless model a single nonzero shot proves that the window is not a desert.
3. **Full coverage gives a deterministic nonzero output.** If every odd offset is covered ($h = 0$) then $\beta(x) = x \bmod 2$ and the output is the fixed string with only the least significant bit set.
4. **The screen is blind to sparse violations.** A single uncovered odd offset gives $h = 2/N$ and probability $h^2 = 4/N^2$, exponentially small in α . More generally, u uncovered odd offsets give probability $(2u/N)^2$. A window with polynomially many uncovered integers therefore looks exactly like a fully covered one to any polynomial number of shots, and Stage 1 cannot verify BGC on a window.
5. **The screen only sees odd offsets.** The value of $S(c+x)$ at even offsets never enters β . This is harmless for macroscopic screening because covered integers occur in runs: an SGP at q covers every integer $y \leq q$ with $y + \lceil 8 \ln y \rceil^2 \geq q$, a run of length about $\lceil 8 \ln q \rceil^2$. Any covered run inside the window contains an odd offset except in degenerate edge cases (for instance a lone covered integer at offset 0). Stage 2 counts all offsets, so the tester's final output is not affected.

Stage 1 has no quantum advantage

Consider a classical screen that evaluates S at R randomly chosen offsets and flags the window if any sampled integer is uncovered. On a desert every sample is uncovered, and on a fully covered window none is. For a window with a small fraction of uncovered integers, one classical sample lands on an uncovered integer with probability about h , while one Deutsch–Jozsa shot returns a zero outcome with probability only h^2 . Per oracle call the classical screen is therefore at least as sensitive, and strictly more sensitive to sparse holes. Under the promise that S is constant on a window, a single classical evaluation already decides which extreme holds. We therefore claim no

quantum advantage for Stage 1; we present it in its Deutsch–Jozsa form because that is the screen this paper is built around, and Section 6 notes that it may be replaced by classical sampling without changing any guarantee about deserts.

5 Stage 2: Quantum Counting Refinement

Let $f(x) = S(c + x)$ for $x \in \{0, \dots, N - 1\}$ and $M = \#\{x : f(x) = 1\}$, the number of covered integers in the window. Stage 2 estimates M by quantum counting (amplitude estimation, Brassard–Høyer–Mosca–Tapp).

- **Cost.** Distinguishing $M = 0$ from $M \geq 1$ takes $\Theta(\sqrt{N}) = \Theta(2^{\alpha/2})$ oracle calls, each costing T_S . The $\Omega(\sqrt{N})$ lower bound holds already for distinguishing $M = 0$ from $M = 1$.
- **Error.** Quantum counting is a bounded-error algorithm, not a deterministic one. With $t = \Theta(\sqrt{N})$ iterations the estimate resolves $M = 0$ from $M \geq 1$ with probability at least $8/\pi^2$, and repetition drives the failure probability below any δ using $O(\log(1/\delta))$ runs. In the standard analysis an empty window is reported as empty with certainty ($M = 0 \Rightarrow \tilde{M} = 0$), so a positive estimate is reliable, and the only error is a nonempty window reported as empty, with probability at most δ .
- **Classical baseline** can be improved using the ideas introduced in this paper as follows. Measured in primality tests, certifying that a window contains no covered point takes about N tests classically: one must in effect rule out an SGP at every integer of the window, and grouping the tests into blocks of $\lceil 8 \ln y \rceil^2$ (one evaluation of S per block) does not reduce the total. In the black-box model for Q' this is an $\Omega(N)$ lower bound, and we know of no sublinear classical method. Counted in evaluations of S alone the classical cost is only about $N/\lceil 8 \ln c \rceil^2$, so we state the comparison at the level of primality tests. Quantum counting on S uses $O(\sqrt{N})$ evaluations, i.e. $O(\sqrt{N} \log^2 c)$ primality tests, and Grover search directly on Q' uses $O(\sqrt{N})$. The advantage is therefore quadratic up to polylogarithmic factors, and it lives in the counting stage. When M is large, a classical algorithm finds a covered point in

about N/M queries and the quantum algorithm in about $\sqrt{N/M}$, so the ratio is again quadratic.

Detecting isolated violations. Stage 1 cannot see a single uncovered integer. The same counting routine applied to the indicator of *uncovered* integers decides whether $U(c, \alpha) = 0$, that is, whether BGC itself holds on the window, again with $\Theta(\sqrt{N})$ oracle calls in the worst case ($U = 1$) against $\Omega(N)$ classically. This is the route to verifying BGC on a window; we do not include it in the tester below because the tester targets density collapse.

6 The Fixed-Interval Tester

Algorithm

Input: start point c , size parameter α , shot budget R , failure probability δ .

1. **Stage 1 (screen).** Run the circuit of Section 4 R times on $W(c, \alpha)$. If all R shots are nonzero, output DENSE and stop.
2. **Stage 2 (refine).** Otherwise the window is *flagged*. Run bounded-error quantum counting on $f(x) = S(c + x)$ with failure probability at most δ , obtaining an estimate $\tilde{M}$. If $\tilde{M} = 0$, output EMPTY; otherwise output SPARSE together with $\tilde{M}$.

We take $R = \text{poly}(\alpha)$, chosen as in part (d) of Proposition 2.

Proposition 2 (guarantees, noiseless model). Let $N = 2^\alpha$ and let h be as in Proposition 1.

- (a) *Deserts are always flagged and detected.* If $W(c, \alpha)$ is a desert then Stage 2 is reached with certainty and the output is EMPTY (with probability at least $1 - \delta$; in fact with certainty when $M = 0$ in the standard analysis).
- (b) *DENSE is a certificate of non-emptiness.* If the output is DENSE, the window is not a desert, with certainty.
- (c) *Compliant windows almost never trigger Stage 2.* If $W(c, \alpha)$ satisfies BGC-Macroscopic, the probability that it is flagged is at most $R h^2 \leq R q(\alpha)^2 / 4^{\alpha-1}$, which is exponentially small in α .

- (d) *Macroscopic deficits are flagged.* If $h \geq \eta$ then the window is flagged with probability at least $1 - (1 - \eta^2)^R \geq 1 - e^{-R\eta^2}$. Taking $R = \lceil \ln(1/\delta)/\eta^2 \rceil$ gives probability at least $1 - \delta$; for $\eta = 1/\text{poly}(\alpha)$ this R is $\text{poly}(\alpha) \ln(1/\delta)$.
- (e) *Cost.* Stage 1 costs R oracle calls, that is RT_S operations. Stage 2, if reached, costs $O(\sqrt{N} \log(1/\delta))$ oracle calls, that is $O(2^{\alpha/2} \log(1/\delta) T_S)$.

Proof. (a) and (b) are consequences 1 and 2 of Proposition 1 together with the Stage 2 analysis of Section 5. (c) A window with $U \leq q(\alpha)$ has $u \leq q(\alpha)$, so $h = u/(N/2) \leq q(\alpha)/2^{\alpha-1}$, and the flag probability is $1 - (1 - h^2)^R \leq Rh^2$. (d) The probability that all R shots are nonzero is $(1 - h^2)^R \leq (1 - \eta^2)^R \leq e^{-R\eta^2}$. (e) is the cost of R oracle evaluations, and the cost stated in Section 5. ■

What the tester does not decide. Windows with $q(\alpha)/2^{\alpha-1} \ll h \ll 1/\sqrt{R}$ may pass the screen with substantial probability, although they violate BGC-Macroscopic as stated. The tester is therefore a one-sided detector of macroscopic collapse, not a decision procedure for BGC-Macroscopic and not a verifier of BGC. Its rigorous, unconditional content is (a)–(e), and only that: it never misses a desert, and it resolves flagged windows into empty or nonempty, for any window whatsoever. Part (c), that it rarely spends quantum counting; is a theorem about windows that satisfy BGC-Macroscopic, not about windows in general; it says nothing about whether the windows this tester will actually meet, sweeping the number line at some given depth, are windows of that kind. Whether they are is exactly the heuristic, unproven question of Section 1, supported there only by a sieve computation up to about $2 \cdot 10^7$ and by a Cramér-type density argument, not by anything established in this section. So the practical claim “Stage 2 will rarely run” is empirical, resting on that evidence, and should not be read as a consequence of (a)–(e). This blind spot does not undermine the tester’s central purpose. What the tester is built to find is a total desert; a run of at least N SGP-free integers, the signature that would speak against the SGP conjecture; and part (a) guarantees a desert is never missed regardless of the blind spot above; a Dense output only ever means “not a total desert,” never a certificate of BGC-Macroscopic. Should a desert ever be flagged and confirmed empty by Stage 2, that would be striking empirical evidence bearing on the SGP conjecture, but establishing that it reflects genuine thinning of the sequence, rather than an isolated large gap consistent

with infinitude, would require further investigation beyond what any single window can show.

Sweeping the number line

To cover a long range, run the tester on contiguous windows $W(c_0 + jN, \alpha)$, $j = 0, 1, \dots, K - 1$, sequentially or in parallel. The look-ahead of S overlaps neighbouring windows by at most $\lceil 8 \ln y \rceil^2 \ll N$ integers and does not affect the analysis. By a union bound, the probability that any of the K Stage 2 decisions is wrong is at most $K\delta$; this is a statement about the reliability of the measurements only, not a probability attached to any conjecture.

Reading the outputs as empirical evidence

- DENSE: the window is certified not to be a desert, and the outcome is what BGC-Macroscopic predicts for an ordinary window. It is not a certificate of BGC.
- SPARSE: an SGP exists within reach of the window, but the screen was triggered, which for a window satisfying BGC-Macroscopic would be an exponentially unlikely event by part (c). It is therefore a candidate counterexample to BGC-Macroscopic, indicating a macroscopic coverage deficit. It is also evidence that SGPs continue to occur at this depth.
- EMPTY: no covered integer exists in the window; there is a run of at least $N = 2^\alpha$ integers with no SGP. This contradicts BGC and BGC-Macroscopic. Under the Cramér-type model of Section 1 the probability of a gap of this length is about $e^{-N/(0.76 \ln^2 c)}$, so it would be an extraordinary event, and would be empirical evidence that the SGP sequence has thinned out at that depth.

No finite computation can prove infinitude or finiteness. An EMPTY output does not prove that SGPs stop: arbitrarily long finite gaps are consistent with infinitude. Conversely DENSE and SPARSE outputs are empirical evidence, of the same kind as classical numerical verification, that SGPs continue to occur. We attach no probability to the conjecture.

7 What Is and Is Not Claimed

1. **Stage 1 is not a quantum speedup.** Classical random sampling matches or exceeds its power (Section 4). The Deutsch–Jozsa circuit is retained as the screen the pipeline is built around, and may be replaced by classical sampling without changing part (a) of Proposition 2.
2. **The quantum advantage is quadratic and lives in Stage 2.** It is the $\Theta(2^{\alpha/2})$ versus $\Omega(2^\alpha)$ separation in certifying that a flagged window is empty (black-box query model, worst case). It is not exponential, and it does not apply to the screening of ordinary windows.
3. **Average-case behaviour.** *If* BGC-Macroscopic holds at the windows actually swept, almost every one of them passes Stage 1 at cost RT_S and Stage 2 is essentially never run (Proposition 2(c)); this is a proven consequence of that hypothesis, not of the tester alone. Whether BGC-Macroscopic in fact holds that deep is the heuristic, empirically-supported claim of Section 1 (checked directly only to about $2 \cdot 10^7$), not something this paper proves. When it does hold, the speed of sweeping a range is due to the screen, which is not quantum, not to the counting stage.
4. **Scope.** The tester detects density collapse. Isolated violations of BGC are invisible to Stage 1 and require the counting routine of Section 5 applied to the uncovered indicator on every window.
5. **Heuristic constants.** The constants 8 and 2 in the bound, and the polynomial q in BGC-Macroscopic, are empirical choices supported by computation up to $2 \cdot 10^7$; Section 1 explains why the universal statement is heuristically expected to fail at very large scale.
6. **Noise.** All guarantees are for the ideal, noiseless circuit. With noisy hardware a lone nonzero shot is no longer a certificate, and the decision rules would need to be re-derived.

8 Some Gains of the Quantum Screen

Sections 4 and 7 established that Stage 1 has no oracle-query advantage over classical sampling. That comparison counts queries to S and says nothing

about two other kinds of value a quantum implementation of Stage 1 can have. We record both explicitly; neither retracts the no-advantage finding above.

A. Coherent reuse across the stage boundary

Remark. Let $T = \{x \in \{0, \dots, N-1\} : x \text{ odd and } S(c+x) = 1\}$, the set of covered odd offsets. The oracle step of the Stage 1 circuit, applied to the uniform superposition with the ancilla in $|-\rangle$, produces (dropping the unentangled ancilla)

$$|\Phi_2\rangle = \frac{1}{\sqrt{N}} \left(\sum_{x \notin T} |x\rangle - \sum_{x \in T} |x\rangle \right), \quad (19)$$

exactly the state obtained by applying a single Grover phase oracle marking T to the uniform superposition. This is immediate from the proof of Proposition 1: $(-1)^{\beta(x)} = -1$ on T and $+1$ elsewhere.

The oracle call already made in Stage 1 is therefore not wasted when a window is flagged: it can be counted as the first query of an amplitude-estimation procedure for $|T|$, rather than discarded after a computational-basis measurement, with a fresh uniform superposition then re-prepared for Stage 2. Concretely, if Stage 2 is implemented as Iterative Quantum Amplitude Estimation (Grinko, Gacon, Zoufal, and Woerner, *Iterative Quantum Amplitude Estimation*, npj Quantum Information **7**, 52 (2021)), which builds its estimate entirely from Grover iterations on the marking oracle with no separate phase-estimation register, Stage 1’s screening rounds and Stage 2’s estimation rounds become rounds of the *same* iterative procedure: early rounds serve as the cheap screen, and only a flagged window continues, coherently, into further rounds, with no intermediate full measurement of the offset register and no re-initialization of the uniform superposition at the stage boundary.

We are careful about what this does and does not do. It removes one classical-readout-and-restart boundary and reuses one oracle call: a constant, architectural saving, not a change to the $O(\sqrt{N})$ versus $\Omega(N)$ separation of Section 5, which counts total oracle calls and is unaffected by how those calls are grouped into circuits. It also targets $|T|$ (covered odd offsets), not $M = \#\{x : S(c+x) = 1\}$ (all covered offsets); Section 4, consequence 5, already argues these track together for the tester’s purposes, since a covered

run of length $\Theta(\log^2 y)$ almost never consists entirely of even offsets, so using $|T|$ in place of M throughout does not change any claim made about the tester elsewhere in this paper.

B. Value independent of query advantage

Not every quantum construction needs a query-complexity advantage to be worth reporting for example as illustrated in the 2013 paper (Locate ref) with the quantum algorithm for number theory there.

In the same sense, the boosted oracle of Section 2 and the Stage 1 circuit map the covering structure of the Bounded Gap Conjecture onto a Deutsch–Jozsa-type circuit, exhibiting the SGP-covering indicator as an amplitude that can be manipulated coherently (Section A above) rather than only sampled. That this particular circuit does not beat classical sampling at its stated task (Section 4) does not remove this second kind of value; the value is conceptual and architectural rather than a speed claim, and we state it as such rather than letting Section 4’s finding imply the construction has no purpose.

9 Conclusion

We gave a two-stage fixed-interval tester for the Macroscopic Bounded Gap Conjecture. A repeated boosted Deutsch–Jozsa screen has single-shot zero probability exactly h^2 (Proposition 1), which makes it a one-sided detector of macroscopic density collapse that never misses a desert and almost never fires on windows satisfying BGC-Macroscopic, provided the window in fact satisfies it; it is blind to sparse violations and offers no advantage over classical sampling. Every flagged window is passed to quantum counting, which decides emptiness in $O(2^{\alpha/2})$ oracle calls against $\Omega(2^\alpha)$ classically. This quadratic speedup is the quantum contribution of the tester, and the outputs it produces are empirical evidence about the bounded-gap conjecture at the tested depth. The efficiency of the two-stage fixed-interval tester is based on the BCG conjecture being true, check up to 2 million, which allows a modified D-J oracle to act as a cheap filter thus saving resources/time instead of using the more expensive quantum counting sub-algorithm.

A remark on the exponent. The heuristic of Section 1 gives a reason to expect the covering bound used throughout this paper, of order $\ln^2 x$, to eventually fail: under the Hardy–Littlewood density model, the maximal gap up to X grows like $0.76 \ln^2 X \cdot \ln(X/\ln^2 X)$, which is order $\ln^3 X$ to leading order, and so eventually overtakes any fixed-coefficient bound of order $\ln^2 x$; heuristically, near $X \sim 10^{36}$ for the constants used here. A bound of order $\ln^3 x$ instead (for instance $\lceil 8 \ln x \rceil^3$ in place of $\lceil 8 \ln x \rceil^2$) matches the leading order of that same heuristic maximal-gap growth, and the same computation that predicts failure of the $\ln^2$ bound predicts no such failure, at any scale, for a bound of this higher order. We record this to note that we are aware of it, not to adopt it: the oracle construction of Section 2 remains polylog-feasible under either exponent (an $O(\log^3 y)$ rather than $O(\log^2 y)$ chain), so nothing in the tester’s architecture depends on which is used, but we have not repeated the sieve verification of Section 1 for the cubic form, and the heuristic argument for its indefinite survival is exactly as unproven as the argument, given here, for the eventual failure of the quadratic one.